

Supplementary Material — Part A

Taxonomy

*What Commentators Talk About: A Validated Thematic Taxonomy of Event-Grounded Sports
Commentary*

The 34 validated clusters, each with its domain assignment, four-axis outcome, verdict band, lexical signature and representative sentences.

Sections in this part

S1 Cluster Validation Cards (34 Clusters)

S1 — Cluster Validation Cards (34 Clusters)

This supplement provides a per-cluster validation card for each of the 34 clusters in the released taxonomy. Cards are sorted by verdict band (strict-core first) and by cluster size within each band. Each card reports: domain, size, four-axis pass/fail indicators with raw scores, top R3 enriched terms, and three representative sentences. Axis criteria: Algorithm $\geq 4/7$ pipelines surviving (Jaccard ≥ 0.30); Bootstrap mean Jaccard ≥ 0.50 ; Held-out replication rate ≥ 0.50 ; Hierarchical Ward Jaccard ≥ 0.50 .

C6 — Grand Prix wins

Domain: Narrative & meta

Size: 465 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.852 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.686 mean Jaccard

Top R3 enriched terms: grand, grand prix, prix, wins, wins grand, lead grand, british, british grand, win, place grand, dutch, dutch grand, prix victory, lead, prix underway

Representative sentences:

1. *The Oscar goes to the top step of the podium, Oscar Piastri wins the Azerbaijan Grand Prix, and it will be a sequel to celebrate, the Australian is a grand prix winner for the second time.*
2. *The Mexican fans celebrating in that grandstand for Verstappen, who wins the Saudi Arabian Grand Prix with another drive of distinction.*
3. *To a man who's never won in Italy before, and who wins the Emilia Romagna Grand Prix this afternoon, Max Verstappen for the first time in his career takes victory in Italy.*

C0 — DRS-assisted battles

Domain: Race dynamics

Size: 205 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.967 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.820 mean Jaccard

Top R3 enriched terms: drs, got drs, drs zone, got, got, zone, open, drs enabled, enabled, going drs, bit drs, drs going, extra, drs open, drs time

Representative sentences:

1. *He's not got DRS, Fernando Alonso, to help his battle, Gasly has, giving him an extra speed advantage.*
2. *Whether he's going to be as close as Lando Norris was, let's see, he's got DRS in operation, he's getting that extra straight-line speed as they head down towards Stowe corner.*
3. *Now we've got DRS enabled, say goodbye to the Red Bull, that is absolutely outrageous, in another league when he used DRS.*

C31 — Race leader changes

Domain: Race dynamics

Size: 162 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.635 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.606 mean Jaccard

Top R3 enriched terms: race, lead race, lead, race leader, leader, takes lead, race lead, start race, started race, win race, going lead, race going, race control, lap race, race far

Representative sentences:

1. *In comes the race leader, Verstappen also being called in as well, so Verstappen absolutely now in contention for victory.*
2. *Well everyone has been lapped so far by the race leader down to Valtteri Bottas.*
3. *Race leader with a swap on, trying to take avoiding action and he clobbers the bollard out.*

C9 — Formula 1 milestones

Domain: Narrative & meta

Size: 134 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.797 mean Jaccard
✓	Held-out	0.846 replication rate
✓	Hierarchical	0.671 mean Jaccard

Top R3 enriched terms: formula, formula world, time formula, podium formula, podium, formula career, world championship, world, championship, career, best formula, formula 2025, formula podium, 2025, time

Representative sentences:

1. *Welcome back to the Formula 1 World Championship in 2025.*
2. *A huge moment in the world championship, and it's Piastri, Verstappen, and on for his first podium in Formula 1 is Isack Hadjar.*
3. *But that is now half time officially in the Formula One world championship.*

C21 — Braking & lockups

Domain: Race dynamics

Size: 128 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.579 mean Jaccard
✓	Held-out	0.600 replication rate
✓	Hierarchical	0.532 mean Jaccard

Top R3 enriched terms: braking, brakes, braking zone, zone, brake, late, late braking, brake pedal, pedal, lock, brakes just, way traffic, outside brakes, curb, brakes late

Representative sentences:

1. *He just brake-tested me, I got brake-tested like right before the braking zone, it looked like he actually went offline and ended up off the racing.*
2. *Laid on the brakes for Carlos Sainz, late-braking move and a position gained for Sainz who was distraught with being impeded when he had Isack Hadjar in the way yesterday.*
3. *Ocon is ahead going into the braking zone and Vettel locking up can't do anything to stop going into the rear.*

C16 — Safety car deployment

Domain: Regulatory

Size: 122 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.689 mean Jaccard
✓	Held-out	0.562 replication rate
✓	Hierarchical	0.626 mean Jaccard

Top R3 enriched terms: safety, safety car, car, virtual, virtual safety, deployed, car deployed, car safety, ending, car ending, going safety, got safety, lapped cars, timing, car coming

Representative sentences:

1. *But it was a virtual safety car deployed, and Charles Leclerc comes into the pits.*
2. *The virtual safety car is now deployed, that was always coming, and everyone reduces their pace.*
3. *The virtual safety car has been deployed, so Leclerc's coming in, and Mercedes are waiting for George Russell.*

C1 — Time penalties

Domain: Regulatory

Size: 118 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.966 mean Jaccard
✓	Held-out	0.926 replication rate
✓	Hierarchical	0.798 mean Jaccard

Top R3 enriched terms: penalty, time penalty, second penalty, second, second time, 10 second, causing collision, penalty causing, collision, causing, 10, time, penalty forcing, got second, forcing

Representative sentences:

1. *Alonso's been given a 10-second time penalty for causing a collision with Zhou Guanyu.*
2. *Charles Leclerc has been given a five second time penalty for leaving the track and gaining an advantage.*
3. *Kevin Magnussen has been given a 10-second time penalty for leaving the track and gaining an advantage.*

C8 — Race control flags

Domain: Regulatory

Size: 118 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.772 mean Jaccard
✓	Held-out	0.605 replication rate
✓	Hierarchical	0.729 mean Jaccard

Top R3 enriched terms: flag, checkered, checkered flag, yellow, green, yellow flag, chequered, chequered flag, green flag, red flag, red, flag flies, sees checkered, sees, nation driver

Representative sentences:

1. *This is the final corner of the last lap, the Dutch driver looks up and sees a checkered flag first.*
2. *Charles Leclerc started second in the race today, he sees the checkered flag and for the first time other than when he's on pole he takes victory.*
3. *As for the first time he starts a race in Montreal on pole, and for the first time he sees the checkered flag.*

C28 — Corner battles

Domain: Race dynamics

Size: 82 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.617 mean Jaccard
✓	Held-out	0.500 replication rate
✓	Hierarchical	0.675 mean Jaccard

Top R3 enriched terms: turn, run turn, abreast, way turn, run, heading turn, outside turn

Representative sentences:

1. *Lewis Hamilton's there alongside him, now he's on his back foot, now he's going to run wide through turn three.*
2. *Alonso is going to get the run down to turn number four, this is going to be super aggressive between the two of them.*
3. *Do we go three wide into turn number one?*

C22 — Wheel-to-wheel

Domain: Race dynamics

Size: 71 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.711 mean Jaccard
✓	Held-out	0.556 replication rate
✓	Hierarchical	0.639 mean Jaccard

Top R3 enriched terms: wheel, wheel wheel, going wheel, wheel turn, wheel goes, goes wheel, went wheel, wheel action, wheel gets, wheels, went

Representative sentences:

1. *A Red Bull and a Mercedes going wheel to wheel into turn four, Perez stays within the white lines.*
2. *There's Sergio Perez and Charles Leclerc going wheel-to-wheel as they go round turn two.*
3. *Does look good if you're Pierre Gasly, that's for sure, he's going wheel to wheel with Lance Stroll into turn one.*

C10 — Chicane moves

Domain: Race dynamics

Size: 62 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.853 mean Jaccard
✓	Held-out	1.000 replication rate
✓	Hierarchical	0.823 mean Jaccard

Top R3 enriched terms: chicane, second chicane, going chicane, goes chicane, braking, bold, chicane going, make chicane

Representative sentences:

1. *Now there's a lot of momentum, but again, no opportunity, so trying the different lines once more, but it's going to have to be a bold move into the second chicane.*
2. *And that's going to continue all the way down to the braking zone in the second chicane, great view from on board.*
3. *We're heading to the Roggia, and this is the second chicane on the racetrack on the second lap, and Stroll adjudged to be forced off the road.*

C20 — Wing & car damage

Domain: Strategy & technical

Size: 61 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.626 mean Jaccard
✓	Held-out	0.625 replication rate
✓	Hierarchical	0.670 mean Jaccard

Top R3 enriched terms: wing, puncture, damage, damaged, got puncture, contact, bit contact, damage wing, damaged wing, rear, wing endplate, think, endplate, ripped, rear wing

Representative sentences:

1. *Yeah I think they touched actually, there was definitely a little jump from Piastri's car, and that would indicate to me, and it looks like the front wing endplate might be a little bit damaged on Lando.*
2. *Game on once again, this time it's the outside line, oh they make contact, he's got a puncture, he's damaged his front wing.*
3. *Yeah, Norris has a damaged front wing endplate where he's had that brush with Piastri.*

C11 — Team radio celebrations

Domain: Narrative & meta

Size: 55 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.640 mean Jaccard
✓	Held-out	0.700 replication rate
✓	Hierarchical	0.769 mean Jaccard

Top R3 enriched terms: thank, job, mate, weekend, guys, amazing, fantastic, thanks, unbelievable, great job, thank guys, drive, good job, yeah, awesome

Representative sentences:

1. *Thank you very much guys, amazing strategy, we were really really sharp today, and nice pushing 72 laps here, I love it, great job guys.*
2. *What up Max, fantastic job, get her mate, hold on, yeah thanks guys that was a, I think a very good performance, very nice, good race, well executed, good pit stops.*
3. *Nice job guys, well done, excellent result for the team, fantastic, absolutely fantastic, really really amazing drive, that last thing was on fire, well done mate.*

C2 — Lights-out race starts

Domain: Race dynamics

Size: 54 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.876 mean Jaccard
✓	Held-out	0.600 replication rate
✓	Hierarchical	0.703 mean Jaccard

Top R3 enriched terms: lights away, lights, away, away gets, away getting, start, lights lights, away ahead, start good, getting away, better start, gets away, away comes, reaction, better

Representative sentences:

1. *It's lights out and away we go and Max Verstappen gets the better start from Fernando Alonso.*
2. *It's lights out and away we go, Verstappen gets the better start, Perez and Lando Norris jockeying for position.*
3. *It's lights out and away we go and I think Verstappen got a slightly better start than Lewis Hamilton, and also Charles Leclerc a better start than Valtteri Bottas.*

C3 — Mirror checks defending

Domain: Race dynamics

Size: 47 sentences

Verdict band: Strict-core (4/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.532 mean Jaccard
✓	Held-out	0.562 replication rate
✓	Hierarchical	0.728 mean Jaccard

Top R3 enriched terms: mirrors, mirror, checking, scarlet, checks, checking mirrors, checking left, hand mirror, left mirror, looking mirrors, left, checks mirrors, looking, checks mirror, driving mirrors

Representative sentences:

1. *There's a real opportunity now for Max Verstappen, and Charles Leclerc knows it, driving on the mirrors now, checking the left one, checking the right one, trying to keep the world champion behind.*
2. *P7, Tsunoda now checking his mirrors rather than looking ahead.*
3. *But Lando Norris was checking his mirrors a few corners ago, now he sees the crowd rise to applaud him as the winner.*

C27 — Overtake battles

Domain: Race dynamics

Size: 762 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.688 mean Jaccard
✓	Held-out	0.556 replication rate
×	Hierarchical	0.339 mean Jaccard

Top R3 enriched terms: inside, door, corner, gets away, past, outside, open door, getaway, space, ahead, close, goes corner, leads corner, past outside, goes

Representative sentences:

1. *Max Verstappen on the inside has made a brilliant start but then goes off the track as Fernando Alonso closes the door on him.*
2. *Inside line for Lando Norris, roaring around the outside, will they make the corner?*
3. *Charles Leclerc tried to go past Lewis Hamilton, outside line to the inside line, got to be brave to make that work.*

C18 — Team dynamics

Domain: Narrative & meta

Size: 600 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.709 mean Jaccard
✓	Held-out	1.000 replication rate
×	Hierarchical	0.365 mean Jaccard

Top R3 enriched terms: teammate, team, ahead teammate, ahead, past

Representative sentences:

1. *And there's the move for Daniel Ricciardo getting past his old teammate at turn five.*
2. *It's as you were ahead, here we go, the best chance to get past his old teammate.*
3. *This is Pierre Gasly on Yuki Tsunoda, big move and taking the place away from his old teammate and good friend, just out of the points.*

C15 — Tires & compounds

Domain: Strategy & technical

Size: 379 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.827 mean Jaccard
×	Held-out	0.385 replication rate
✓	Hierarchical	0.575 mean Jaccard

Top R3 enriched terms: tires, tire, tyres, hard, medium, compound, tyre, soft, fresher, compound tire, fresher tires, hard compound, medium tires, hard tires, intermediate

Representative sentences:

1. *So on the medium tires that are fresher as Valtteri Bottas is hard compound tire, and with DRS as well he gains on the Mercedes.*
2. *Hard tires going on to Lewis Hamilton's Mercedes, they've not got any new sets of mediums.*
3. *Yes yes everyone's kept a set of new hard compound tires just for the middle phase of this race.*

C14 — Pit stops

Domain: Strategy & technical

Size: 185 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.776 mean Jaccard
✓	Held-out	0.667 replication rate
×	Hierarchical	0.447 mean Jaccard

Top R3 enriched terms: pit, pit stop, lane, pit lane, stop, pits, coming pit, comes pit, come pits, comes, coming, come pit, pitted, stops, slow

Representative sentences:

1. *And my word, Lance Stroll is making a pit stop, something is happening in the pit lane in front of us.*
2. *He is a contender to win this one, as Lando Norris comes in to make his pit stop, leaves into the fast lane he goes, 2.3 seconds and Lando Norris lights up the rear tires, goes back to work.*
3. *Hamilton comes in, Verstappen and Perez stay out, Hamilton it's a slow stop again but he had to wait for the other car that was coming down the pit lane.*

C13 — Championship & title

Domain: Narrative & meta

Size: 158 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.693 mean Jaccard
✓	Held-out	0.600 replication rate
×	Hierarchical	0.492 mean Jaccard

Top R3 enriched terms: championship, world, world championship, champion, world champion, title, constructors, win, time world, championship leader, trophy, 2025, points, leader, 2024

Representative sentences:

1. *The world championship leader up against the two-time world champion highly regarded Fernando Alonso.*
2. *The world championship leader Oscar Piastri didn't finish, but he still leads by 25 points.*
3. *As we go on to the penultimate tour, Piastri who leads the championship by nine points trying to score win number six of the World Championship.*

C12 — Podium finishes

Domain: Narrative & meta

Size: 139 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
✓	Bootstrap	0.826 mean Jaccard
×	Held-out	0.462 replication rate
✓	Hierarchical	0.767 mean Jaccard

Top R3 enriched terms: podium, podium places, home, place podium, comes home, podium place, completes, places, finish, completes podium, podium finish, complete podium, step, step podium, stand podium

Representative sentences:

1. *Max Verstappen completes the podium with third place, and Charles Leclerc in his own race comes home to finish, for the first time.*
2. *Sergio Perez provided great support to his teammate and to his team to come home and take the final podium place as well.*
3. *As Charles Leclerc comes home for his first podium finish of the season.*

C30 — Lap pace & timing

Domain: Strategy & technical

Size: 136 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.429 survival fraction
✓	Bootstrap	0.697 mean Jaccard
✓	Held-out	0.900 replication rate
✓	Hierarchical	0.534 mean Jaccard

Top R3 enriched terms: lap, fastest lap, fastest, opening, previous lap, opening lap, previous, lap lap, turn lap, lap goes, lap going, lap time, quicker, end lap, middle

Representative sentences:

1. *Perez in clear air, finally away from the back markers, 1:19.468 new fastest lap, gap down to six seconds.*
2. *And he might just be slowing down a bit as he crosses the line, he did slow down, he did abort the fastest lap attempt.*
3. *There's the championship points at the moment, Hamilton picking up 19 because of that fastest lap.*

C33 — Field position changes

Domain: Race dynamics

Size: 122 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.571 survival fraction
✓	Bootstrap	0.510 mean Jaccard
✓	Held-out	0.619 replication rate
×	Hierarchical	0.392 mean Jaccard

Top R3 enriched terms: seventh, place, sixth, fifth, sixth place, eighth, seventh place, 11th, position, fifth place, sixth position, 10th, ninth, 12th, seventh position

Representative sentences:

1. *Norris is up to ninth, P10 for Lewis Hamilton, Russell versus Albon, this is for sixth position, or it was, because Albon has just driven around the side of his good friend and taken sixth place.*
2. *He moves up to sixth position, Perez down to seventh.*
3. *Oscar Piastri still in fifth position, no change for anyone until you look all the way back down to Lewis Hamilton who's up to seventh position, gaining a couple of places, as well as Hülkenberg.*

C19 — Race lead holding

Domain: Race dynamics

Size: 115 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.857 survival fraction
✓	Bootstrap	0.694 mean Jaccard
×	Held-out	0.300 replication rate
✓	Hierarchical	0.620 mean Jaccard

Top R3 enriched terms: lead, leads, leads second, lead second, maintains lead, holds lead, lead comes, holds, leads comes, maintains, leads inside, leads leads, takes lead, maintain lead

Representative sentences:

1. *Lando Norris leads, yeah that was a great start from him, I think there was some contact we saw a little puff of smoke but no visible damage, and uh yeah what a great great start for Lando Norris.*
2. *Hamilton leads from Verstappen, then comes Sergio Perez and then the Ferrari of Charles Leclerc.*
3. *Leclerc leads Verstappen on the inside, Sainz on the outside, Verstappen stays ahead going into turn one.*

C5 — Weather conditions

Domain: Strategy & technical

Size: 43 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
×	Bootstrap	0.075 mean Jaccard
✓	Held-out	0.667 replication rate
✓	Hierarchical	0.716 mean Jaccard

Top R3 enriched terms: rain, rain coming, wet, minutes, rain falling, falling, light, expected, track, extraordinary, class, lasting, temperature, 10 minutes, 30

Representative sentences:

1. *Coming out of turn three, the light rain falling, and the seven-time world champion has found the barriers.*
2. *Rain expected in seven laps lasting for 10 minutes, and if it's heavy rain, 10 minutes of it is going to really change things.*
3. *Yeah it is just that light rain, it's still falling, but it's not enough now, and I think the teams are getting more confidence that it's going to pass without significantly dampening the track.*

C4 — Race start launch

Domain: Race dynamics

Size: 42 sentences

Verdict band: Robust (3/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
×	Bootstrap	0.358 mean Jaccard
✓	Held-out	0.600 replication rate
✓	Hierarchical	0.861 mean Jaccard

Top R3 enriched terms: launch, launch line, good launch, line, good, got good, got launch, gets launch, chops, pole sitter, sitter, chop, got, initial launch, great launch

Representative sentences:

1. *A good launch off the line for Lando Norris, sees him chop across Charles Leclerc, but Charles got a great amount of momentum but was blocked.*
2. *Good launch off the line for our pole-sitter Oscar Piastri, but also George Russell as well who's alongside Max Verstappen.*
3. *Verstappen's got a good launch, he's already up to second, I reckon that's the race lead in the inside line.*

C26 — Defending & attacking

Domain: Race dynamics

Size: 110 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	0.714 survival fraction
✓	Bootstrap	0.664 mean Jaccard
×	Held-out	0.333 replication rate
×	Hierarchical	0.481 mean Jaccard

Top R3 enriched terms: fight, defend, battle, defensive, going fight, try defend, defended, try fight, defense, fights, going

Representative sentences:

1. *Fernando Alonso knew what was coming, he fights his way by, good battle between these two, now resolved in favor of Alonso.*
2. *And now Stroll up against Ocon is the battle we're looking at, and Ocon's going to fight away by, now these two get on pretty well, they still get on brilliantly.*
3. *Perez fights it with one hand tied behind his back a little bit, he wasn't going to put up too much defense.*

C23 — Circuit action

Domain: Race dynamics

Size: 72 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
✓	Bootstrap	0.700 mean Jaccard
×	Held-out	0.400 replication rate
✓	Hierarchical	0.585 mean Jaccard

Top R3 enriched terms: racetrack, corner racetrack, turn racetrack, turns racetrack

Representative sentences:

1. *Oliver Bearman, Hamilton clears Max Verstappen, no fireworks compared to what we've seen before from these two, notably around this racetrack in 2021.*
2. *Piastrri on a different part of the racetrack, it is dirty and dusty off the racing line, driving on his mirrors now the rookie Australian.*
3. *And Piastrri going very brave to a dirty dusty part of the racetrack to make the move into turn 23.*

C17 — Gap & pace

Domain: Strategy & technical

Size: 67 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
✓	Algorithm	1.000 survival fraction
×	Bootstrap	0.461 mean Jaccard
×	Held-out	0.333 replication rate
✓	Hierarchical	0.602 mean Jaccard

Top R3 enriched terms: seconds, gap, tenths, hour, miles hour, tenths second, gap seconds, miles, half, extra, 20, pace, 10 seconds, second, 10

Representative sentences:

1. *The gap at the front, yeah, 5 and a half seconds, Lando doing his best lap of the race, but Oscar's holding that pace firm, it's tight, it's tight.*
2. *And don't forget the gap was about one and a half seconds, point seven it was yeah, between Verstappen and Hamilton.*
3. *We certainly are, 20.6 for Hamilton, that's his pace, 1.2 seconds taking out of him, that gap down to six seconds.*

C7 — Race start lights

Domain: Race dynamics

Size: 36 sentences

Verdict band: Moderate (2/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.429 survival fraction
×	Bootstrap	0.041 mean Jaccard
✓	Held-out	0.636 replication rate
✓	Hierarchical	0.864 mean Jaccard

Top R3 enriched terms: lights, red lights, prix underway, underway, grand, grand prix, prix, red, lights time, lights away, green flag, begins, racing lights, green, away

Representative sentences:

1. *The green flag at the back tells us we go racing in five lights time and the Austrian Grand Prix is underway.*
2. *One, two, three, four, five red lights in Budapest, and the Hungarian Grand Prix is underway.*
3. *Two, three, four, five red lights in Budapest, and the Hungarian Grand Prix is underway.*

C29 — Corner specific moves

Domain: Race dynamics

Size: 60 sentences

Verdict band: Weak (1/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
×	Bootstrap	0.445 mean Jaccard
×	Held-out	0.375 replication rate
✓	Hierarchical	0.717 mean Jaccard

Top R3 enriched terms: turn, turn 11, 11, turn 10, turn seven, 13, turn 12, 12, inside turn, seven, heading turn, heading, coming turn, turn 13, 10

Representative sentences:

1. *We'll send it to turn seven, high speed stuff all the time.*
2. *But let's have a look once again at this part, because now we're on the short burst up towards turn seven.*
3. *Nico Hülkenberg following into turn seven and is, no, I wondered if that was the moment.*

C25 — Driver & team references

Domain: Narrative & meta

Size: 69 sentences

Verdict band: Fragile (0/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
×	Bootstrap	0.472 mean Jaccard
×	Held-out	0.455 replication rate
×	Hierarchical	0.481 mean Jaccard

Top R3 enriched terms: driver, nation driver, nation, drivers, driver getting, board nation, driver trying, contract

Representative sentences:

1. *McLaren started this weekend by announcing they've got both drivers signed up long term, Oscar Piastri has a bumper contract, and he has the race leader close ahead of him.*
2. *So Yuki doesn't love that idea, we're talking about working with a senior team, they got to work with their own drivers at the moment.*
3. *And in we come at the same time, McLaren and Mercedes elect to pit their drivers, Antonelli and Piastri and Hamilton all in on the same lap.*

C24 — Inter-car distance

Domain: Strategy & technical

Size: 60 sentences

Verdict band: Fragile (0/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.143 survival fraction
×	Bootstrap	0.411 mean Jaccard
×	Held-out	0.417 replication rate
×	Hierarchical	0.475 mean Jaccard

Top R3 enriched terms: car, cars, car length, car lengths, length, lengths, couple car, width, got car, length ahead, car ahead, car width, flat floor, floor, flat

Representative sentences:

1. *And this is the closest yet Perez has been, yes it is, and he's only got a couple of car lengths.*
2. *And he's got a quarter of a car length in front now, they come out of turn two and he's got a couple of car lengths in front.*
3. *Alonso makes his way out of the first corner and he's only a couple of car lengths behind Leclerc.*

C32 — Front position changes

Domain: Race dynamics

Size: 45 sentences

Verdict band: Fragile (0/4)

Four-axis validation:

	Axis	Score
×	Algorithm	0.000 survival fraction
×	Bootstrap	0.241 mean Jaccard
×	Held-out	replication rate
×	Hierarchical	0.407 mean Jaccard

Top R3 enriched terms: fourth, place, position, fourth place, second, second place, couple positions

Representative sentences:

1. *Up to fourth place goes Pierre Gasly, down a couple of positions goes Kevin Magnussen.*
2. *Charles Leclerc comes out in fourth place.*
3. *Fourth place now for Carlos Sainz.*